

Underactuated Humanoid Locomotion

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Learning-based control method for bipedal walking

Abstract—This brief describes a simulation-first approach to humanoid walking and underactuated motion. A published walking controller provides a baseline for local policy development through imitation and reinforcement learning. Physical diagnostics and visual inspection guide bounded experiments. The immediate task is recognizable walking; deliberately passive joints follow.

Index Terms—Humanoid locomotion, underactuation, simulation, reinforcement learning.

I. MODEL AND EXECUTE THE DYNAMICS

Use a free base, actuated legs, and fixed upper-body joints. For configuration q and generalized velocity v , the dynamics are [1]

$$\begin{aligned} M(q)\dot{v} + h(q, v) &= B\tau + J_c^\top \lambda, \\ \dot{q} &= N(q)v. \end{aligned} \quad (1)$$

M is inertia, h includes gravity and velocity terms, B maps motor torques τ , $J_c^\top$ maps contact forces λ , and N handles orientation kinematics. The base is unactuated; contact changes control authority. Fixed flat-foot support can be fully actuated in reduced coordinates.

Native C++ connects MuJoCo contact and friction dynamics to LibTorch learning and inference. Physics runs on the CPU; neural computation uses CUDA. A browser shows live simulation frames and motion/contact diagnostics.

II. ESTABLISH THE WALKING BASELINE

Replay the published Unitree controller [2] with its documented observation and action interface. Preserve it as a reference alongside a zero-torque control. The local PPO actor receives command c and causal history H of angular velocity, projected gravity, joint states, and prior actions:

$$\begin{aligned} a_t &\sim \pi_\theta(\cdot \mid H_t, c_t), \\ q_{\text{des}} &= q_{\text{nom}} + D \tanh(a_t), \\ \tau &= \text{sat}[K_p(q_{\text{des}} - q_a) - K_d v_a]. \end{aligned} \quad (2)$$

θ denotes policy parameters and D bounds nominal-pose offsets. Joint states q_a , v_a drive faster proportional-derivative (PD) feedback; sat enforces motor limits. Preserve preprocessing and action transformations when exporting and evaluating each policy.

An asymmetric critic receives additional simulator state during training. The actor uses observable history without privileged critic signals.

III. DEVELOP AND ADAPT LOCAL POLICIES

Imitation trains a student from the reference controller's action labels. Independent reinforcement learning starts from random weights without teacher labels. Proximal policy optimization (PPO) [3] collects parallel rollouts, estimates advantages, and applies clipped updates to optimize

$$\max_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]. \quad (3)$$

T is the horizon and γ the discount factor. Fit return targets and bootstrap at nonterminal truncations. Reward forward tracking, uprightness, and alternating support; penalize falls, slip, effort, and abrupt actions. Revise rewards using physical diagnostics: contact events alone need not represent useful steps.

Use an existing walker as a starting point for adaptation. Record whether a candidate is pretrained, teacher-assisted, trained from random weights, or resumed; these support different learning claims.

IV. VERIFY MOTION WITHIN A FIXED BUDGET

Fix the task and budget before training. Freeze candidate weights and evaluate continuous flat-ground trials from independently reset initial conditions. Stop on a fall or the horizon without automatic resets. Inspect foot lifting, load transfer, alternation, displacement, and slip alongside rendered motion. Distinguish walking from pivoting, sliding, or repeated same-leg lifts.

Archive configurations, checkpoint hashes, training histories, and evaluations. Stop at the criterion or budget; further training requires a separate decision. Finite demonstrations do not establish general stability.

V. EXTEND TOWARD PASSIVE-JOINT MOTION

Once walking is established, compare the unchanged reference with a movable joint whose motor torque is disabled. Verify actual actuator effort. Adaptation can then target compensation through remaining joints and contact dynamics. Hardware transfer and broader robustness remain future work.

REFERENCES

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